01 / Claim

What capability are we testing, under which conditions?

Mode: Source analysis / Project development Claim: _____ Source/configuration freeze: _____ Date: _____

Our [system] will [perform a capability] in [conditions], evaluated using [measure and criterion] against [comparison].

1. System and capability

Name your system, tested capability and physical mechanism. Separate your proposed claim from published results.

2. Conditions and boundary

State task, configuration, conditions and exclusions. Use Project development for your own final project.

3. Measure and comparison

Define measure, units, trial unit, criterion and comparator before evaluation. Label proposed versus observed values.

4. Purpose and next test

Select one feasible pilot and check resources/access. State what outcome would challenge the claim; results may be pending.

Review: Could your system work as shown while your claim is still false? Write one such outcome.

Worksheet 1 / Claim - evidence and team review

Class 12 | Due 13 November 2026 before 10:50 | Include every canvas and review page

Team / project: _____ Claim ID: _____ Worksheet revision: _____

Date: _____ Mode: Project development (label source analysis separately) Pages / IDs: _____

Configuration / evidence version (an earlier commit or artifact ID): _____

Evidence ID	Repo-relative path + evidence ID	Origin; what it supports; limitation / access gap

Missing evidence or next action / owner / verification:

Each member signs after reviewing the reasoning, evidence labels, limitations and next action. Signatures apply to all identified pages; they do not certify success or readiness. Do not sign for another member.

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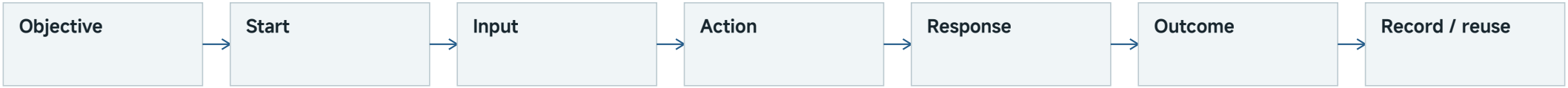
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Push one legible complete PDF to submissions/class12/worksheet.pdf in your assigned private team repository by the deadline. TA collects this file; no physical hand-in. Include identified continuations; keep later revisions separate.

02 / Experience

What interaction data does one episode generate, and how will we use them?

Mode: Source analysis / Project development Claim: _____ Source/configuration freeze: _____ Date: _____



1. Objective and start

Link the claim, episode start/reset and end rule. Label this episode planned, team-observed or externally reported.

2. Input and action

Identify observations, timestamps, commands and executed actions where known. Distinguish command from response.

3. Response and outcome

Record response, outcome rule, failures, timeouts and interventions. A selected success does not establish reliability.

4. Record and reuse

List the retained interaction data: observations, actions, responses, outcomes and context. State intended use and missing fields.

Review: What could make the recorded outcome misleading? Mark missing measurements or an alternative explanation.

Worksheet 2 / Experience - evidence and team review

Class 13 | Due 17 November 2026 before 10:20 | Include every canvas and review page

Team / project: _____ Claim ID: _____ Worksheet revision: _____

Date: _____ Mode: Project development (label source analysis separately) Pages / IDs: _____

Configuration / evidence version (an earlier commit or artifact ID): _____

Evidence ID	Repo-relative path + evidence ID	Origin; what it supports; limitation / access gap

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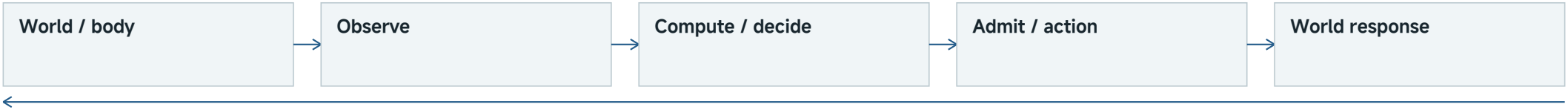
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03 / System

What system produces that interaction?

Mode: Source analysis / Project development Claim: _____ Source/configuration freeze: _____ Date: _____



Feedback, operator enable/stop/recovery and evidence recording cross the loop.

1. World and observations

Draw embodiment/environment and sensing. Mark frames, measured quantities and implemented versus planned components.

2. Decision and action

Connect observations to computation/control and action. Declare action meaning, units, order, rates and limits.

3. Authority and record

Add operator authority, logging, stop/reset/recovery and configuration ID. Identify evidence kept at each critical interface.

4. Critical interface evidence

Select one consequential interface. State acceptance criterion, test evidence and result, or pending test with owner.

Review: If an interface is late, missing or misordered, how will you detect it and what action is allowed?

Worksheet 3 / System - evidence and team review

Class 14 | Due 18 November 2026 before 11:50 | Include every canvas and review page

Team / project: _____ Claim ID: _____ Worksheet revision: _____

Date: _____ Mode: Project development (label source analysis separately) Pages / IDs: _____

Configuration / evidence version (an earlier commit or artifact ID): _____

Evidence ID	Repo-relative path + evidence ID	Origin; what it supports; limitation / access gap

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Push one legible complete PDF to submissions/class14/worksheet.pdf in your assigned private team repository by the deadline. TA collects this file; no physical hand-in. Include identified continuations; keep later revisions separate.

04 / Grounding

What connects the representation or interpretation to physical evidence?

Mode: Source analysis / Project development Claim: _____ Source/configuration freeze: _____ Date: _____

Prediction / representation	Matched reference	Difference / ambiguity	Conclusion / scope limit

1. Comparison boundary

Name the prediction/measurement and reference. Match conditions; identify simulated versus physical evidence.

2. Quantities and validity

Declare quantity, units, frame, tolerance and measurement limits. Distinguish a chosen tolerance from observed uncertainty.

3. Results and uncertainty

Give paired outcomes, trial counts and uncertainty or limits. If untested, record the protocol and missing evidence explicitly.

4. Interpretation and challenge

Bound the conclusion and consider another cause of the difference. Choose a diagnostic test or scope restriction.

Review: Could sensing error, reset differences or selection explain the discrepancy instead of model error?

Worksheet 4 / Grounding - evidence and team review

Class 15 | Due 20 November 2026 before 10:50 | Include every canvas and review page

Team / project: _____ Claim ID: _____ Worksheet revision: _____

Date: _____ Mode: Project development (label source analysis separately) Pages / IDs: _____

Configuration / evidence version (an earlier commit or artifact ID): _____

Evidence ID	Repo-relative path + evidence ID	Origin; what it supports; limitation / access gap

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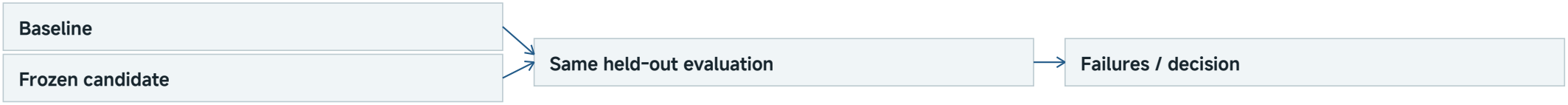
Signature: _____ Date: _____

Push one legible complete PDF to submissions/class15/worksheet.pdf in your assigned private team repository by the deadline. TA collects this file; no physical hand-in. Include identified continuations; keep later revisions separate.

05 / Learning

How is experience used, and what contribution is supported?

Mode: Source analysis / Project development Claim: _____ Source/configuration freeze: _____ Date: _____



1. Role of experience

State how interaction data support training, inference, evaluation or design. No new training is required; explain your scope.

2. Data and independence

Identify data origin, selection and development/test groups. Keep related episodes together; disclose tuning or leakage.

3. Comparison and limits

Compare frozen methods or data strategies under matched conditions and budgets. Include denominators and all failure outcomes.

4. Supported contribution

Retain, revise or reject the bounded contribution. No improvement is valid; an unfinished evaluation is not a negative result.

Review: Would the apparent improvement survive the same reset, action limits and trial budget for both methods?

Worksheet 5 / Learning - evidence and team review

Class 16 | Due 24 November 2026 before 10:20 | Include every canvas and review page

Team / project: _____ Claim ID: _____ Worksheet revision: _____

Date: _____ Mode: Project development (label source analysis separately) Pages / IDs: _____

Configuration / evidence version (an earlier commit or artifact ID): _____

Evidence ID	Repo-relative path + evidence ID	Origin; what it supports; limitation / access gap

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06 / Decision – full 1-3-9-27 screen

What bounded conclusion or next action does the evidence justify?

Mode: Source analysis / Project development Claim: _____ Source/configuration freeze: _____ Date: _____

Mode / claim or next action: _____ Source/configuration freeze: _____ Scope: _____

Generate grounded experience

G1 / Grounded physical problem

G1.1 Mechanism: Is the claimed mechanism explicit, physically credible and open to challenge?

G1.2 Closed loop: Are the implemented body, observations, controller, actions, environment and human roles identified?

G1.3 Task boundary: Are task conditions, episode boundaries, outcomes and exclusions stated?

G2 / Measurable embodiment

G2.1 Embodiment: Is the embodiment or physical coupling realized and identified at the claimed stage?

G2.2 Observability: Are measurements and references suitable for the claimed outcomes, with relevant validity limits?

G2.3 Interfaces: Are consequential interfaces, action meanings, units and timing described or verified?

G3 / Controlled physical evidence

G3.1 Commissioning: What limits, resets, interventions, fault handling and recovery are described or tested?

G3.2 Experience record: What observations, actions, outcomes and contextual records are available or described?

G3.3 Grounding gap: How are model, measurement or interpretation connected to physical evidence under comparable conditions?

Use experience for claims

U1 / Bounded claim logic

U1.1 Claim: Can the available evidence support or challenge the specific claim, without expanding its scope?

U1.2 Role of experience: Is the role of experience in learning, inference, adaptation, evaluation or design insight explicit?

U1.3 Comparators: Are comparators or alternative explanations appropriate to the contribution being claimed?

U2 / Representative experience

U2.1 Correspondence: Are data, model, policy, embodiment and evaluation configurations linked where needed?

U2.2 Coverage: Are coverage, sampling, failures, evaluation partitions and leakage addressed for the claim?

U2.3 Governance: What provenance, rights, human participation and data safeguards are reported where relevant?

U3 / Reproducibility and challenge

U3.1 Reconstruction: What is available to reconstruct the system and evaluation, and what remains unverified?

U3.2 Evaluation: Are protocol, denominators, repetitions, uncertainty and exclusions reported appropriately?

U3.3 Failure analysis: Are failure modes, interventions, competing explanations and negative results examined?

Manage evidence through lifecycle

M1 / Evidence traceability

M1.1 Identification: Are claims, configurations, datasets and experiments distinguishable and source-linked?

M1.2 Change control: Are variants, updates or corrections traceable, and is their effect on conclusions assessed?

M1.3 Counterevidence: Are material failures, null findings, ambiguous outcomes and counterevidence retained?

M2 / Resources, burdens and benefits

M2.1 Resources: Are relevant hardware, compute, time, skills and maintenance resources stated?

M2.2 Data burden: Are collection, annotation, reset, storage and human effort addressed where consequential?

M2.3 Benefit: Is the claimed benefit justified against costs and relevant alternatives within the study?

M3 / Continuing operation

M3.1 Monitoring: What detects mismatch, degradation, invalid measurements or faults in the claimed setting?

M3.2 Operations: What human authority, maintenance, recovery and update practices are described for this stage?

M3.3 Sustainment: What duration, units, contexts or repeated-use evidence supports the claimed persistence?

Support (strength for the bounded question)

0 contradicted; 1 weak; 2 partial; 3 adequate within scope; 4 convergent. NR not reported in inspected scope; NE not examined; NA not applicable with a reason.

Evidence context (separate from origin, support and access)

E1 reasoning/design/plan; E2 executed analysis/simulation/digital artifact; E3 physical/integrated experiments, including externally reported results; E4 specified physical coverage/auditability; E5 independent reproduction/sustained operation within scope. Not a universal quality ladder. NR/NE/NA: context normally a dash.

06 / Decision – evidence and action record

Cite source evidence; state findings and limits. Apply action gates only to project development.

Mode: Source analysis / Project development Claim: _____ Source/configuration freeze: _____ Date: _____

Mode / claim or next action / source scope: _____

G1 / Grounded physical problem

Check	Evidence ID + basis	Support	Context
G1.1			
G1.2			
G1.3			

State: _____ Gap/NA rationale: _____
Action / owner / verification: _____

G2 / Measurable embodiment

Check	Evidence ID + basis	Support	Context
G2.1			
G2.2			
G2.3			

State: _____ Gap/NA rationale: _____
Action / owner / verification: _____

G3 / Controlled physical evidence

Check	Evidence ID + basis	Support	Context
G3.1			
G3.2			
G3.3			

State: _____ Gap/NA rationale: _____
Action / owner / verification: _____

U1 / Bounded claim logic

Check	Evidence ID + basis	Support	Context
U1.1			
U1.2			
U1.3			

State: _____ Gap/NA rationale: _____
Action / owner / verification: _____

U2 / Representative experience

Check	Evidence ID + basis	Support	Context
U2.1			
U2.2			
U2.3			

State: _____ Gap/NA rationale: _____
Action / owner / verification: _____

U3 / Reproducibility and challenge

Check	Evidence ID + basis	Support	Context
U3.1			
U3.2			
U3.3			

State: _____ Gap/NA rationale: _____
Action / owner / verification: _____

M1 / Evidence traceability

Check	Evidence ID + basis	Support	Context
M1.1			
M1.2			
M1.3			

State: _____ Gap/NA rationale: _____
Action / owner / verification: _____

M2 / Resources, burdens and benefits

Check	Evidence ID + basis	Support	Context
M2.1			
M2.2			
M2.3			

State: _____ Gap/NA rationale: _____
Action / owner / verification: _____

M3 / Continuing operation

Check	Evidence ID + basis	Support	Context
M3.1			
M3.2			
M3.3			

State: _____ Gap/NA rationale: _____
Action / owner / verification: _____

Choose the conclusion for the declared mode

Source analysis: summarize supported contributions, partial support, contradictions and unresolved questions by claim family. Distinguish claim-critical checks from extension inquiries. Reporting gaps do not automatically invalidate the contribution. No whole-paper average or readiness award.

Project development only: identify critical checks and requirements for the next action. Green: adequate support, no blocking unknown. Amber: manageable condition. Red: essential condition absent or contradicted. NR/NE cannot satisfy a dependent critical requirement.

Bounded conclusion / unsupported uses: _____

Follow-up inquiry or conditions / owner / verification: _____

Reassessment trigger / invalidated evidence / version: _____

Retain all 27 checks, evidence bases and limitations. Do not turn missing reporting into contradiction or a favorable mean into approval.

Worksheet 6 / Decision - evidence and team review

Class 17 | Due 25 November 2026 before 11:50 | Include every canvas and review page

Team / project: _____ Claim ID: _____ Worksheet revision: _____

Date: _____ Mode: Project development (label source analysis separately) Pages / IDs: _____

Configuration / evidence version (an earlier commit or artifact ID): _____

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